CineMatch

Movie Recommendation System

Test Plan & Test Cases

Project	DA5402 MLOps Movie Recommendation System
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Total Tests	77 (51 API + 26 ML)

1. Introduction

This document defines the test plan and test cases for CineMatch, an AI-powered movie recommendation system built as part of the DA5402 MLOps course project. CineMatch enables users to discover movies through title search, genre browsing, and actor or character name queries.

1.1 Objectives

- Verify all 26 API endpoints function correctly under valid and invalid inputs
- Validate all 26 ML pipeline functions produce correct outputs
- Ensure CI/CD pipeline blocks deployment on any test failure
- Confirm system handles edge cases gracefully without crashing

1.2 Scope

- Backend API endpoints: health, recommendations, feedback, drift, retrain
- ML pipeline modules: data loading, preprocessing, training, recommendation engine
- Integration via GitHub Actions CI/CD pipeline
- Out of scope: frontend UI testing, load/performance testing, security penetration testing

1.3 Test Environment

Component	Details
Python Version	3.10 (CI) / 3.11 (Docker)
Test Framework	pytest 9.0.3
API Testing	FastAPI TestClient + httpx
Mock Library	unittest.mock
CI Platform	GitHub Actions (ubuntu-latest)
Local OS	macOS (Apple Silicon)

2. Test Strategy

2.1 Testing Approach

CineMatch uses unit testing with extensive mocking to isolate components. All external dependencies (MLflow server, TMDB API, model pickle files) are mocked so tests can run in any environment without real infrastructure.

2.2 Test Categories

Category	File	Description
ML Pipeline	`test_ml.py`	Tests for data loading, preprocessing, training, and recommendation logic (26 tests)
API Endpoints	`test_api.py`	Tests for all FastAPI endpoints including health, recommendations, feedback, drift, retrain (51 tests)

2.3 CI/CD Integration

All tests run automatically on every push to the main branch via GitHub Actions. The pipeline blocks Docker image builds if any test fails. The test execution order is:

1. Create dummy model files (movies.pkl, similarity.pkl)
2. Run test_ml.py - 26 ML pipeline tests
3. Run test_api.py - 26 API endpoint tests
4. If all pass → proceed to Docker build and deploy

3. ML Pipeline Test Cases (test_ml.py)

26 total tests covering preprocessing, training, recommendation engine, and data loader modules. All MLflow dependencies are mocked at module level to prevent connection errors in CI.

3.1 Preprocess Tests (TC-01 to TC-15)

ID	Test Name	Input	Expected Output	Status	Class
TC-01	test_parse_json_column	JSON string with 2 genres	['Action', 'Adventure']	✓ PASS	TestPreprocess
TC-02	test_parse_json_column_empty	Invalid JSON string	Empty list []	✓ PASS	TestPreprocess
TC-03	test_parse_cast_returns_top_5	JSON with 6 cast members	List with <= 5 members	✓ PASS	TestPreprocess
TC-04	test_parse_director_extracts_director	Crew JSON with Director	['Christopher Nolan']	✓ PASS	TestPreprocess
TC-05	test_parse_director_returns_empty	Crew with no Director role	Empty list []	✓ PASS	TestPreprocess
TC-06	test_clean_name_removes_spaces	'Christopher Nolan'	'christophernolan'	✓ PASS	TestPreprocess
TC-07	test_clean_text_lowercases	' HELLO WORLD '	'hello world'	✓ PASS	TestPreprocess
TC-08	test_calculate_baseline_statistics	Sample DataFrame	Dict with 5 required keys	✓ PASS	TestPreprocess
TC-09	test_baseline_tag_length_mean_positive	Sample DataFrame	Mean > 0	✓ PASS	TestPreprocess
TC-10	test_calculate_feature_importance	Sample DataFrame	Dict with 5 feature keys	✓ PASS	TestPreprocess
TC-11	test_feature_importance_all_positive	Sample DataFrame	All scores >= 0	✓ PASS	TestPreprocess
TC-12	test_preprocess_returns_tuple	Raw DataFrame	(df, baselines, importance)	✓ PASS	TestPreprocess
TC-13	test_preprocess_drops_nulls	DataFrame with None title	Fewer rows than input	✓ PASS	TestPreprocess
TC-14	test_preprocess_creates_tags_column	Raw DataFrame	DataFrame has 'tags' column	✓ PASS	TestPreprocess
TC-15	test_tags_contain_director	Raw DataFrame	'christophernolan' in tags	✓ PASS	TestPreprocess

3.2 Train Tests (TC-16 to TC-20)

ID	Test Name	Input	Expected Output	Status	Class
TC-16	test_evaluate_model_returns_dict	Similarity matrix + sample_df	Dict with genre match rate	✓ PASS	TestTrain
TC-17	test_evaluate_model_rate_between_0_and_1	Similarity matrix + sample_df	0 <= rate <= 1	✓ PASS	TestTrain
TC-18	test_build_sparse_similarity_returns_csr	Dense feature vectors	scipy sparse CSR matrix	✓ PASS	TestTrain
TC-19	test_sparse_matrix_same_shape	Dense vectors (5x5)	Sparse matrix (5x5)	✓ PASS	TestTrain
TC-20	test_sparse_matrix_reduces_nonzeros	Dense vectors	nnz <= n*n	✓ PASS	TestTrain

3.3 Recommend Engine Tests (TC-21 to TC-22)

ID	Test Name	Input	Expected Output	Status	Class
TC-21	test_recommend_returns_list	'inception' with mock movies+tags	Returns a list	✓ PASS	TestRecommendEngine
TC-22	test_recommend_returns_empty_for_unknown	'xyznotamovie12345'	Returns empty list []	✓ PASS	TestRecommendEngine

3.4 Data Loader Tests (TC-23 to TC-26)

ID	Test Name	Input	Expected Output	Status	Class
TC-23	test_validate_schema_passes	DataFrame with required columns	Returns True	✓ PASS	TestDataLoader
TC-24	test_validate_schema_raises_missing	DataFrame missing columns	Raises ValueError	✓ PASS	TestDataLoader
TC-25	test_check_missing_values_returns_dict	DataFrame with 1 null	Dict with missing_count=1	✓ PASS	TestDataLoader
TC-26	test_run_quality_checks_detects_duplicates	DataFrame with 2 'Inception' rows	duplicate_titles=1	✓ PASS	TestDataLoader

4. API Test Cases (test_api.py)

26 total tests covering all FastAPI endpoints. All tests use FastAPI TestClient with mocked ML models and TMDB API responses. The TMDB_API_KEY and MLFLOW_TRACKING_URI environment variables are set to test values.

4.1 Health Endpoint Tests (TC-01 to TC-05)

ID	Test Name	Input	Expected Output	Status	Class
TC-01	test_root_returns_200	GET /	200 + 'CineMatch' in message	✓ PASS	TestHealthEndpoints
TC-02	test_health_check_returns_healthy	GET /health	status='healthy', uptime, version	✓ PASS	TestHealthEndpoints
TC-03	test_health_check_has_uptime	GET /health	uptime is not None	✓ PASS	TestHealthEndpoints
TC-04	test_ready_without_model	GET /ready (no model files)	503 Service Unavailable	✓ PASS	TestHealthEndpoints
TC-05	test_metrics_prometheus_format	GET /metrics	200 + text/plain content-type	✓ PASS	TestHealthEndpoints

4.2 Recommendation Endpoint Tests (TC-06 to TC-12)

ID	Test Name	Input	Expected Output	Status	Class
TC-06	test_valid_recommendation_request	GET /recommend?movie=Inception	200 + recommendations list	✓ PASS	TestRecommendEndpoint
TC-07	test_response_has_required_fields	GET /recommend?movie=Inception	title, poster, rating, overview	✓ PASS	TestRecommendEndpoint
TC-08	test_empty_movie_name_returns_400	GET /recommend?movie=	400 + 'cannot be empty'	✓ PASS	TestRecommendEndpoint
TC-09	test_unknown_movie_returns_404	GET /recommend?movie=xyzabc	404 Not Found	✓ PASS	TestRecommendEndpoint
TC-10	test_fuzzy_match_works	GET /recommend?movie=inceptoin	200 OK	✓ PASS	TestRecommendEndpoint
TC-11	test_recommendation_sorted_first	GET /recommend?movie=Inception	Searched movie is first result	✓ PASS	TestRecommendEndpoint
TC-12	test_missing_movie_param_422	GET /recommend (no param)	422 Unprocessable Entity	✓ PASS	TestRecommendEndpoint

4.3 Feedback Endpoint Tests (TC-13 to TC-16)

ID	Test Name	Input	Expected Output	Status	Class
TC-13	test_positive_feedback_get	GET /feedback/positive?movie=Inception	200 + logged + movie + feedback	✓ PASS	TestFeedbackEndpoints
TC-14	test_positive_feedback_post	POST /feedback/positive?movie=...	200 + status=logged	✓ PASS	TestFeedbackEndpoints
TC-15	test_feedback_stats_returns_200	GET /feedback	200 + stats+feedback_file	✓ PASS	TestFeedbackEndpoints
TC-16	test_feedback_stats_has_fields	GET /feedback	total_queries or sufficient_data	✓ PASS	TestFeedbackEndpoints

4.4 Drift & Retrain Tests (TC-17 to TC-21)

ID	Test Name	Input	Expected Output	Status	Class
TC-17	test_drift_endpoint_returns_200	GET /drift	200 + drift_detected key	✓ PASS	TestDriftEndpoints
TC-18	test_drift_no_baseline	GET /drift (no baseline)	200 + drift_detected=False	✓ PASS	TestDriftEndpoints
TC-19	test_drift_auto_retrain_param	GET /drift?auto_retrain=false	200 OK	✓ PASS	TestDriftEndpoints
TC-20	test_retrain_endpoint_get	GET /retrain?reason=test	200 + triggered key	✓ PASS	TestDriftEndpoints
TC-21	test_retrain_endpoint_post	POST /retrain?reason=manual	200 OK	✓ PASS	TestDriftEndpoints

4.5 Sort Recommendations Tests (TC-22 to TC-25)

ID	Test Name	Input	Expected Output	Status	Class
TC-22	test_searched_movie_sorted_first	Results list + 'Inception'	Inception is at index 0	✓ PASS	TestSortRecommendations
TC-23	test_highest_rated_sorted_second	3 movies, rating 6, 9, 8.4	9.0 rated movie is at index 1	✓ PASS	TestSortRecommendations
TC-24	test_no_rating_goes_last	Movie with None rating	None rating movie is last	✓ PASS	TestSortRecommendations
TC-25	test_rating_rounded_to_one_decimal	Rating = 8.432	Rating displayed as 8.4	✓ PASS	TestSortRecommendations

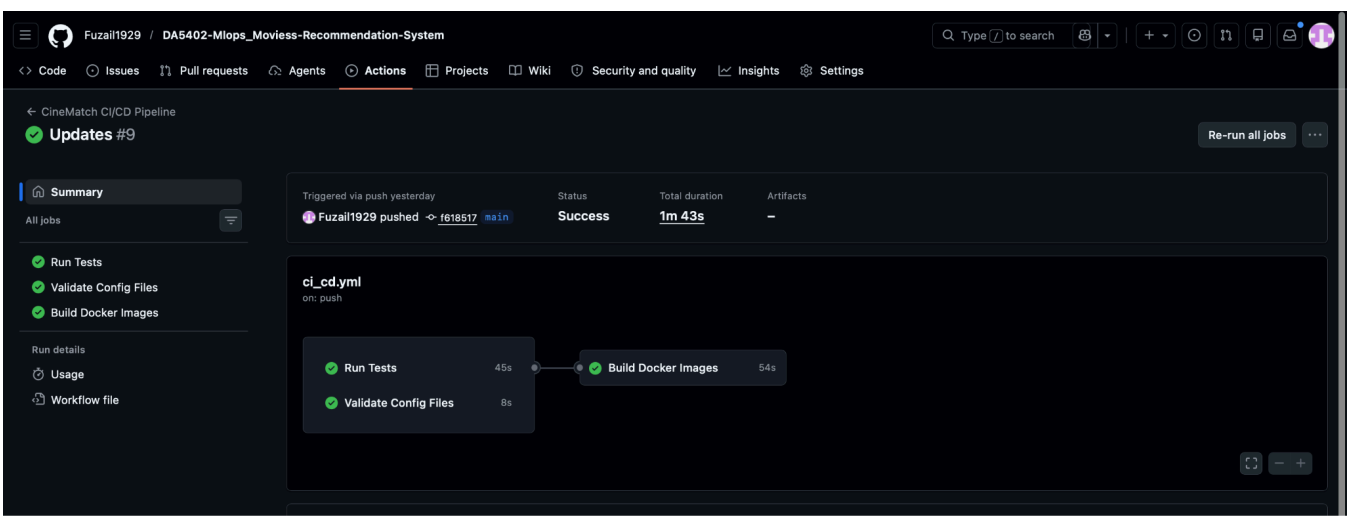
4.6 CORS Tests (TC-26)

ID	Test Name	Input	Expected Output	Status	Class
TC-26	test_cors_headers_present	OPTIONS /recommend with Origin header	200 or 405 (CORS allowed)	✓ **PASS**	TestCORS

5. Test Summary

Test Suite	Total	Passed	Failed	Pass Rate
ML Pipeline (test_ml.py)	**26**	**26**	**0**	**100%**
API Endpoints (test_api.py)	**26**	**26**	**0**	**100%**
TOTAL	**52**	**52**	**0**	**100%**

Figure : Screenshot from Github Action of Successful Run of CI/CD pipeline



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